

Java Script Basics

GXEST203

Foundations of Computing: From Hardware Essentials to Web Design

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Java Script Basics

JavaScript

- JavaScript is a high-level, interpreted programming language.
- It is primarily used for web development.
- Enables dynamic content, interactivity, and event handling on web pages.
- Examples include updating content, controlling multimedia, and animating images.
- JavaScript is mainly used alongside HTML and CSS.
- Provides interactivity to web pages.
- JavaScript can control the Document Object Model (DOM).

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Java Script Basics

How JavaScript Fits into a Web Page

- JavaScript can be embedded directly in HTML.
- External JavaScript files can also be linked.
- JavaScript code is executed by the web browser.
- Manipulate HTML and CSS.
- Respond to user events like clicks and form submissions.
- Handle data and make web pages dynamic.

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Variables and Data Types in Java Script

- Variables store data values.
- Data types in JavaScript:
 - String: Textual data.
 - Number: Numeric data.
 - Boolean: True or False.
 - Object: Complex data structure.

Variables - js file

```
// variables.js

// Declaring a string variable
let name = "John";

// Declaring a number variable
let age = 30;

// Declaring a boolean variable
let isEmployed = true;

// Logging the values of variables to the console
console.log("Name:␣", name);
console.log("Age:␣", age);
console.log("Employed:␣", isEmployed);
```

Variables - html file

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,␣
    initial-scale=1.0">
  <title>Variables Demo</title>
</head>
<body>
  <h1>JavaScript Variables Demo</h1>
  <p>Open the console to see the output.</p>
  <script src="variables.js"></script>
</body>
</html>
```

JS - Variables

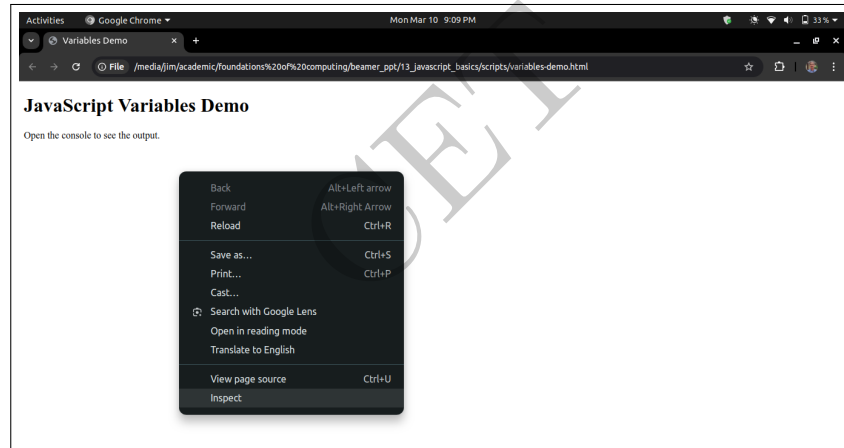


Figure: Opening the console : Right click & Inspect

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JS - Variables

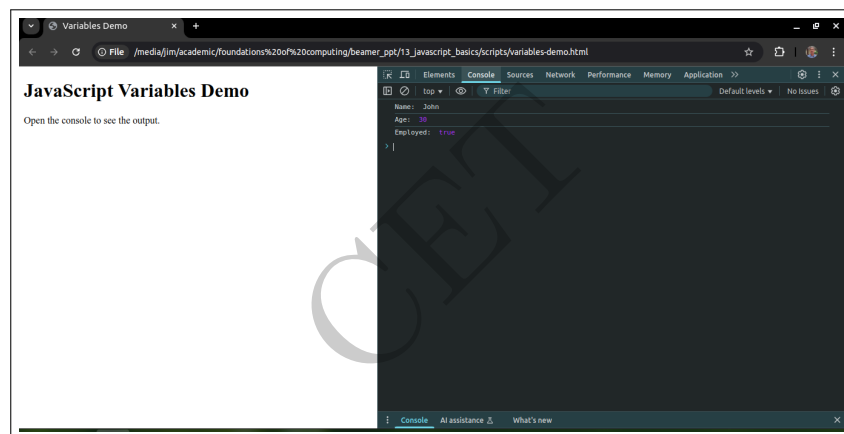


Figure: Console log messages displayed

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Control Structures in JavaScript

Control Structures are used to control the flow of execution

- **Conditional Statements:**

- if statement
- if-else statement
- else if statement
- switch statement

- **Loops:**

- for loop
- for...in loop
- for...of loop
- while loop
- do...while loop

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Control Structures - js file

```
// control-structures.js
// For loop iterates 5 times
for (let i = 0; i < 5; i++) {
    console.log("Iteration:", i); // Prints the current
    iteration number
}
// An if-else control structure to check age
let age = 18;
// Check if the age is 18 or above
if (age >= 18) {
    console.log("You are an adult.");
} else {
    console.log("You are a minor.");
}
```

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Control Structures - html file

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
        initial-scale=1.0">
    <title>Control Structures Demo</title>
</head>
<body>
    <h1>Control Structures Demo</h1>
    <p>Open the console to see the output.</p>
    <script src="control.js"></script>
</body>
</html>
```

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JS - Control Structures

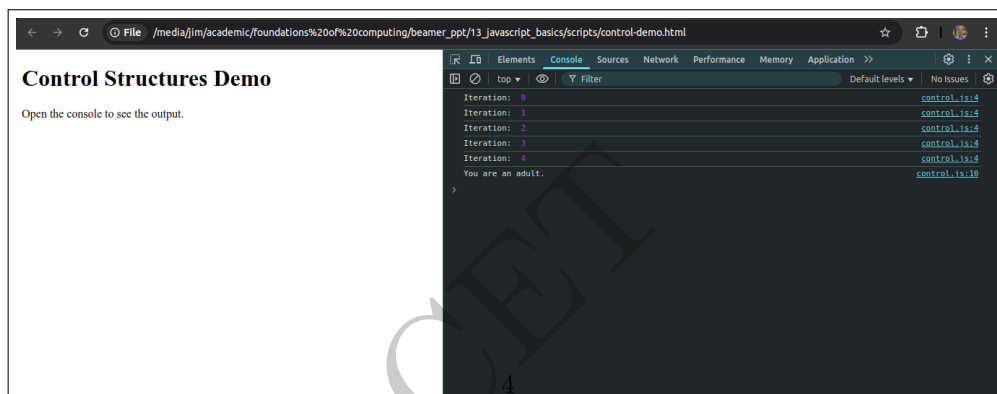


Figure: Console log messages displayed

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Functions - js file

```
// functions.js

// Declaring a function named 'greet' that takes one
// parameter 'name'
function greet(name) {
    // The function returns a greeting message with the
    // provided name
    return "Hello, " + name + "!";
}

// Calling the function with different arguments and logging
// the result
console.log(greet("Alice")); // Output: Hello, Alice!
console.log(greet("Bob"));  // Output: Hello, Bob!
```

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Java Script Basics

Functions - html file

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
        initial-scale=1.0">
    <title>Functions Demo</title>
</head>
<body>
    <h1>Functions Demo</h1>
    <p>Open the console to see the output.</p>
    <script src="functions.js"></script>
</body>
</html>
```

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Java Script Basics

JS - Functions

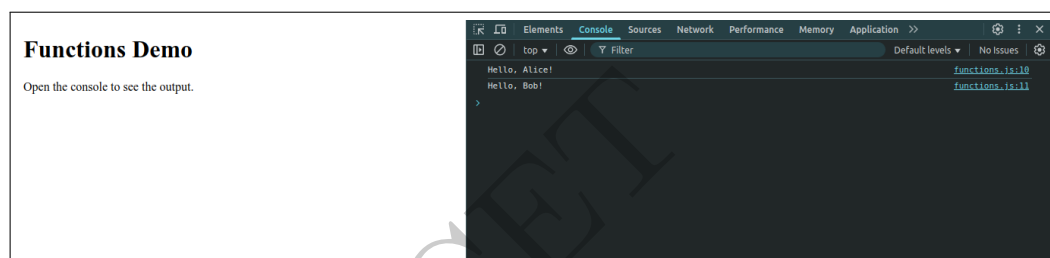


Figure: Console log-messages displayed

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Java Script Basics

Document Object Model (DOM)

- The Document Object Model (DOM) represents the structure of a web page.
- When a web page is loaded, the browser creates a Document Object Model of the page.
- It treats the web page as a tree structure (nodes).
- Through the DOM, JavaScript can modify elements, change styles, add or remove content, and respond to user events.

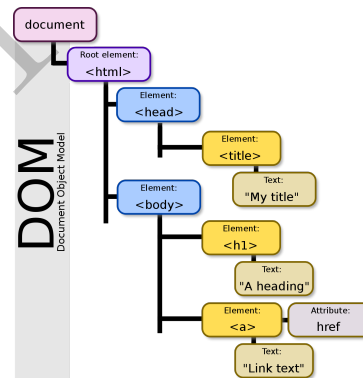


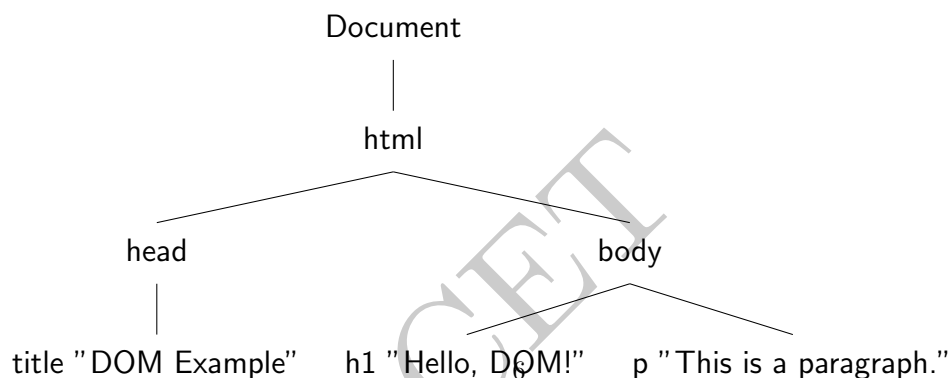
Figure: Structure of DOM

Document Object Model (DOM)

Listing 1: An html page

```
<!DOCTYPE html>
<html>
  <head>
    <title>DOM Example</title>
  </head>
  <body>
    <h1>Hello, DOM!</h1>
    <p>This is a paragraph.</p>
  </body>
</html>
```

Document Object Model (DOM)



DOM of the above html page

Example : Displaying Time with JavaScript - time.js

```
// time.js
// Function to display the current date and time
function displayTime() {
    // Create a new Date object to get the current date and
    // time
    let date = new Date();
    // Find the HTML element with the ID 'time' and update
    // its text content
    document.getElementById('time').innerText = date;
    console.log("Time updated to: " + date)
}
// Continuously update the time every second (1000
// milliseconds)
setInterval(displayTime, 1000);
```

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Java Script Basics

Adding script to the html code of the Web Page

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
        initial-scale=1.0">
    <title>Live Clock</title>
</head>
<body>
    <h2>Current Time</h2>
    <div id="time">Loading...</div>

    <script src="time.js"></script>
</body>
</html>
```

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Java Script Basics

Current time displayed by the web page

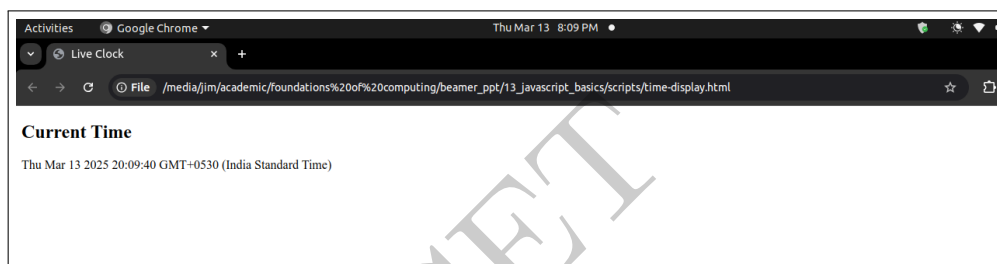


Figure: Current time displayed by the web page

Testing the Script

- Use web browser developer tools.
- Check for errors in the console.
- Verify output and functionality.

Debugging using Chrome DevTools

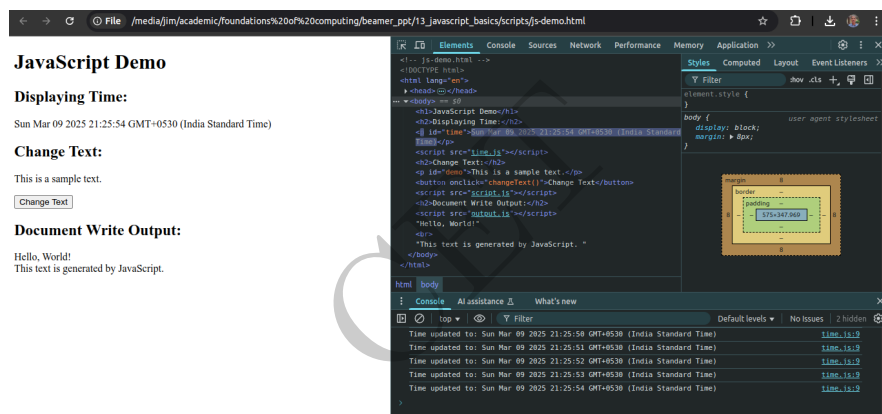


Figure: Debugging time.js using Chrome DevTools

Chrome DevTools is a collection of tools built into Google Chrome Browser that helps you inspect, debug, and improve websites.

DOM Manipulation: Selecting Elements (EXTRA TOPIC)

DOM (Document Object Model) allows JavaScript to manipulate webpage content.

Selecting Elements:

- `getElementById("id")` - Select by ID
- `getElementsByClassName("class")` - Select by class
- `getElementsByTagName("tag")` - Select by tag
- `querySelector("selector")` - Select first match
- `querySelectorAll("selector")` - Select all matches

Modifying Content:

- `textContent` - Change text content
- `innerHTML` - Change HTML content
- `style.property` - Modify style

Selecting by ID

Using `getElementById()`

```
// HTML
<p id="demo">Hello World!</p>

// JavaScript
var element = document.getElementById("demo");
element.style.color = "blue";
```

Selecting by Class

Using `getElementsByClassName()`

```
// HTML
<p class="message">Hello</p>
<p class="message">World!</p>

// JavaScript
var elements = document.getElementsByClassName("message");
for (var i = 0; i < elements.length; i++) {
    elements[i].style.fontWeight = "bold";
}
```

Selecting by Tag Name

Using `getElementsByTagName()`

```
// HTML
<div>
  <p>This is a paragraph.</p>
  <p>This is another paragraph.</p>
</div>

// JavaScript
var paragraphs = document.getElementsByTagName("p");
for (var i = 0; i < paragraphs.length; i++) {
    paragraphs[i].style.color = "green";
}
```

Selecting by Query Selector

Using `querySelector()`

```
// HTML
<div>
  <p class="example">First paragraph</p>
  <p class="example">Second paragraph</p>
</div>

// JavaScript
var element = document.querySelector(".example");
element.style.backgroundColor = "yellow";
```

Selecting Multiple Elements

Using `querySelectorAll()`

```
// HTML
<ul>
  <li>Item 1</li>
  <li>Item 2</li>
  <li>Item 3</li>
</ul>

// JavaScript
var items = document.querySelectorAll("li");
items.forEach(item => item.style.color = "red");
```

DOM manipulation - js file (EXTRA TOPIC)

```
// dom-select-modify.js

// Select element by ID
let heading = document.getElementById("heading");
heading.textContent = "Modified_Heading_using_getElementById";

// Select element by class name
let paragraphs = document.getElementsByClassName("text");
for (let i = 0; i < paragraphs.length; i++) {
  paragraphs[i].style.color = "blue";
}

// Select element by tag name
let listItems = document.getElementsByTagName("li");
for (let i = 0; i < listItems.length; i++) {
  listItems[i].style.fontWeight = "bold";
}

// Select first element by CSS selector
let firstParagraph = document.querySelector(".text");
firstParagraph.style.fontSize = "18px";

// Select all elements by CSS selector
let allItems = document.querySelectorAll("li");
allItems.forEach(item => {
  item.style.textTransform = "uppercase";
});
```

DOM manipulation - html file (EXTRA TOPIC)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>DOM Selection and Modification</title>
</head>
<body>

  <h1 id="heading">Original Heading</h1>

  <p class="text">This is the first paragraph.</p>
  <p class="text">This is the second paragraph.</p>

  <ul>
    <li>Item One</li>
    <li>Item Two</li>
    <li>Item Three</li>
  </ul>

  <script src="dom-select-modify.js"></script>
</body>
</html>
```

JS - DOM manipulation (EXTRA TOPIC)

Modified Heading using getElementById

This is the first paragraph.

This is the second paragraph.

- ITEM ONE
- ITEM TWO
- ITEM THREE

Figure: Webpage

Adding the Script to a Web Page (EXTRA TOPIC)

```
<!-- js-demo.html -->
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>JavaScript Demo</title>
</head>
<body>
  <h1>JavaScript Demo</h1>

  <h2>Displaying Time:</h2>
  <p id="time"></p>
  <script src="time.js"></script>

  <h2>Change Text:</h2>
  <p id="demo">This is a sample text.</p>
  <button onclick="changeText()">Change Text</button>
  <script src="script.js"></script>

  <h2>Document Write Output:</h2>
  <script src="output.js"></script>
</body>
</html>
```

script.js executing

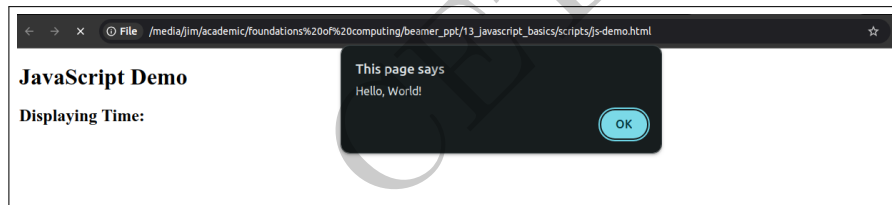


Figure: Alert message upon loading the web page

script.js (EXTRA TOPIC)

```
// script.js
// Display an alert message when the page loads
alert('Hello, World!');

// Function to change the text content of the paragraph
function changeText() {
    // Find the HTML element with the ID 'demo'
    // Update its text content to a new message
    document.getElementById('demo').innerText = 'Text
    changed by JavaScript!';
}
```

script.js executing (EXTRA TOPIC)



Figure: Web page after alert message. time.js is displaying the time

output.js executing (EXTRA TOPIC)



Figure: Web page text changed by script.js

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Write text directly to the HTML document using output.js

```
// output.js
// Write text directly to the HTML document
// This will output "Hello, World!" followed by a line break
document.write('Hello, World!<br>');

// Write another line of text to the HTML document
document.write('This text is generated by JavaScript.');
```

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